

All necessary formulas for quadratics
(You must memorize all of these for SAT.)

When $ax^2 + bx + c = 0$ is given

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum of the solutions

$$-\frac{b}{a}$$

Product of the solutions

$$\frac{c}{a}$$

Vertex form when vertex/min/max is given as (h, k)

$$y = a(x - h)^2 + k$$

To get the a value, you will need to find and plug in an additional point for x and y .

Quadratic regression on desmos (need 3 points)

1. The graph of a quadratic function $y = f(x)$ in the xy -plane intersects the x -axis when $x = -8$ and when $x = p$, where p is a constant. The minimum value of $f(x)$ occurs at the point $(1, m)$, where m is a constant. What is the value of p ?

2. $f(x) = (x - p)(x - q)$

$$f(-2) < 0, f(1) > 0, \text{ and } f(4) < 0$$

If p and q are integers, what 'could' be the value of $p + q$?

3. A quadratic function $f(x)$ has the vertex $(-6, 6)$ and passes through $(1, -2)$. What is the value of $f(-13)$?

4. A quadratic function $g(x) = ax^2 + bx + c$ contains $(0, 5)$, $(1, 4)$, and $(-2, 13)$. What is the value of b ?

5. The product of the solutions to the equation below is jky . What is the value of y ?

$$18x^2 + (18j + k)x - jk = 0$$

6. What is the value of the sum of the solutions to the equation below?

$$2x^2 - 6x + 3 = 0$$